



Graph & Geometric Deep Learning for Drug Toxicity Prediction

Can models that read molecular structure directly beat classical fingerprint baselines at predicting toxicity – and does the answer depend on the task?

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DILI
+0.043
GNN wins
GIN 0.870 vs RF 0.827

CLINTOX
+0.020
GNN wins
GIN 0.820 vs XGB 0.800

AMES
-0.012
Classical wins
RF 0.792 vs GIN 0.780

hERG
-0.018
Classical wins
RF 0.798 vs GINE 0.780

LD50 · REGRESSION
-0.050
Classical leads on paper – but every GNN scores **negative R²** under scaffold splits. A leakage signal, not a real win (Panel 4A).

1 The Study
Four representation families, one fair benchmark

Molecules are graphs: atoms are nodes, bonds are edges. We ask whether learning **directly on that structure** – in 2D or 3D – outperforms hand-crafted fingerprints across toxicity endpoints of differing size and balance.

SMILES Input
CC(=O)Oc1ccccc1
Aspirin

Molecular Graph
Node features + bond connectivity

GNN Layers
Input Layer 1 Layer 2
Message passing aggregates neighbours

Prediction
TOXIC
p = 0.84
NON-TOXIC
p = 0.12

SMILES → molecular graph → message-passing GNN → per-molecule toxicity probability.

CLASSICAL
Random Forest, XGBoost on Morgan fingerprints (ECFP4, 1024-bit)

LANGUAGE
ChemBERTa – frozen RoBERTa pre-trained on 77M SMILES + linear head

GNN · 2D
GCN, GIN, GAT, GraphSAGE, GINE – graph topology; GINE adds bond features

GNN · 3D
SchNet on RDKit ETKDG 3D conformer geometry

Datasets 5 TDC toxicity tasks (DILI, ClinTox, AMES, hERG, LD50) + the 12-endpoint Tox21 panel. Protocol Bemis–Murcko scaffold splits (structurally disjoint train/test), 3 seeds, 70/10/20, class-weighted loss. Metric AUROC, or R² for LD50.

2 Main Result — the winner depends on the task
Best GNN minus best classical AUROC, per dataset

Classical wins

GNN wins

DILI (n=475, 45% pos.)
ClinTox (n=1484, 8% pos.)
AMES (n=7255, 50% pos.)
hERG (n=13445, 40% pos.)
LD50 R² (n=7385, regr.)

-0.06 -0.04 -0.02 0.00 0.02 0.04 0.06

Best GNN AUROC – Best Classical AUROC (scaffold splits, 3 seeds)

Best GNN – best classical AUROC per dataset (scaffold splits, 3 seeds). Bars right = GNN advantage; left = classical advantage.

A consistent split emerges. **GNNs win on the hard cases** – small, imbalanced datasets such as DILI (n=475) and ClinTox (8% positive), where learning structure-aware representations pays off. **Classical fingerprints win on the large, balanced cases** – AMES and hERG – where reactive substructures are common and a fingerprint already captures the decisive signal.

No single method dominates. Architecture choice should follow dataset size, class balance and chemistry – not a default preference for the newest model.

3 Full Benchmark — 8 methods × 5 datasets
AUROC, mean over 3 scaffold-split seeds (R² for LD50) · bold = best per column · † = std > 0.03

		AMES	ClinTox	DILI	hERG	LD50 (R2)
Classical	RF	0.792	0.733	0.827	0.798	0.030
	XGBoost	0.757	0.800	0.815	0.794	-0.049
	ChemBERTa	0.708	0.625	0.701	0.688	—
GNN	GCN	0.718	0.691	0.845	0.692	-0.073
	GIN	0.780	0.820	0.870	0.778	-0.523
	GAT	0.749	0.745	0.836	0.703	-0.020
	GraphSAGE	0.774	0.815	0.840	0.716	-0.009
	GINE	0.758	0.810	0.860	0.780	-0.755

No universal winner. GIN and GINE top the imbalanced classification tasks; RF and XGBoost top the balanced ones.

- ChemBERTa finishes last on every classification dataset despite 77M-molecule pre-training – frozen SMILES features do not transfer to unseen scaffolds.
- GIN tops two endpoints (ClinTox 0.820, DILI 0.870); GINE's bond features add a further edge on DILI (0.860) and hERG.
- Every GNN scores negative R² on LD50 regression under scaffold splits – foreshadowing the leakage finding below.

Pre-training is not a shortcut. Under a realistic generalisation test, a 500-tree Random Forest on fingerprints beats a 77M-parameter language model on all four classification tasks.

4A LD50 — a leakage cautionary tale

Random split Scaffold split

R²

0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00

RF XGB GCN GIN GAT SAGE GINE

collapse

Under random splits GIN looks strong (R²=0.49). Under scaffold splits every GNN collapses below zero – GIN -0.52, GINE -0.76.

Random-split LD50 measures memorisation, not prediction. The widely cited R²=0.49 for GNNs is data leakage.

4B Does 3D geometry help?

SchNet (3D) GINE (2D+bonds)

AUROC

0.85
0.80
0.75
0.70
0.65

ClinTox AMES

0.810 0.773 0.711 0.756

SchNet (3D conformers) never beats GINE (2D + bonds) on any seed; the gap narrows from 0.10 on ClinTox to 0.017 on AMES as data grows.

3D buys uncertainty, not accuracy. Prediction spread across conformers reliably flags unreliable molecules – a free, post-hoc reliability signal.

4C Multi-task Tox21

Multi-task wins Single-task wins

Multi-task – Single-task AUROC

0.10
0.05
0.00
-0.05
-0.10

NR-AR NR-AR-LBD NR-AIR NR-Arom- NR-ER NR-ER-LBD NR-PPAR-Y SR-ARE SR-ATAD5 SR-HSE SR-MMP SR-p33

macro avg

One shared GIN backbone across all 12 Tox21 endpoints vs 12 single-task models. **Macro-average AUROC drops by 0.024**, only a few related receptors (NR-AR-LBD, SR-ATAD5) benefit.

Forced sharing hurts. The 12 pathways are too biologically diverse to share one representation.

★ Conclusions

1 The winner depends on the task. GNNs lead on hard, imbalanced datasets; classical fingerprints lead on large, balanced ones. No single method dominates.

2 LD50 benchmarks can mislead. Scaffold splits expose near-zero generalisation; the widely cited R²=0.49 for GNNs is data leakage.

3 3D geometry buys uncertainty, not accuracy. SchNet trails GINE on every seed, but conformer variance flags unreliable predictions for free.

4 Evaluation protocol changes the story. Multi-task transfer hurts, frozen ChemBERTa underperforms, and random-split results cannot be trusted – scaffold splitting is essential.

REFERENCES

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SETUP & REPRODUCIBILITY

RDKit featurisation · PyTorch Geometric · scikit-learn / XGBoost. All runs use a shared Murcko scaffold splitter for cross-method comparability; per-run weights and metrics are checkpointed for resumability.
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Scan for the interactive microsite
thekoning.github.io/AML_Microsite